

Homework 1

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July 18, 2026

1 Overview

In this project, we were tasked with using the breast cancer dataset to implement and evaluate at least five classifiers from the following list:

- Logistic Regression
- Naive Bayes
- Perceptron
- Support Vector Machine
- Decision Tree Classifier
- K-Nearest Neighbors

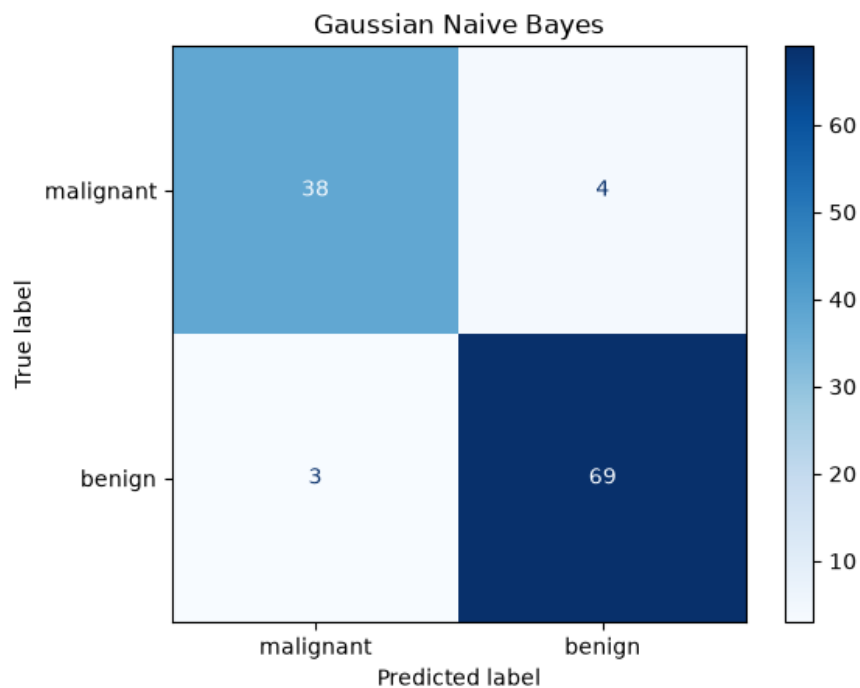
2 Preprocessing

In the processing step, we split the data into 80 percent training and 20 percent testing. When splitting data, we set the stratify parameter to yes. This feature ensures that, in each split, classes that are rare, such as a positive cancer diagnosis, occur with the same frequency as in the main data set. We also opted to test different scalers (MinMax, Standard, no scaler). We did this by using Pipeline and GridSearchCV to test multiple different parameters at once.

3 Helper Functions

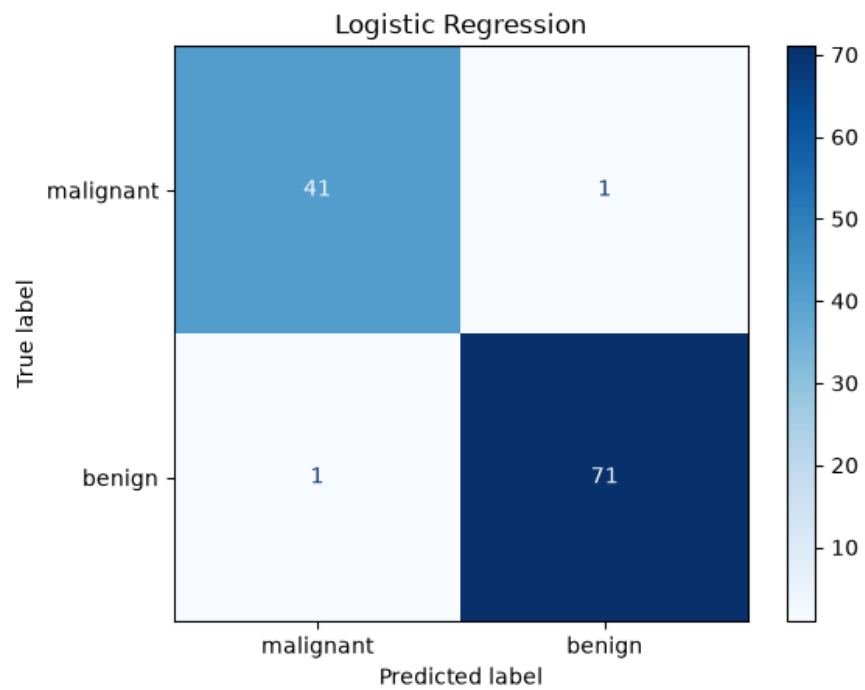
We used three helper functions in our notebook: `evaluate`, `summarize_grid`, and `weighted_recall`. The “`evaluate`” function takes the model and its name as input parameters. The function trains, tests, displays the confusion matrix, prints the classification report, and returns the model metrics. The “`summarize_grid`” function is used to compare different scalers. One of the questions asked how different scaling methods affect different algorithms; this function allows us to take a closer look at this. The “`weighted_recall`” function takes model predictions, true values, the weights of malignant and benign recall, and the weight of benign recall as input parameters. This function then computes a “weighted recall.” Since this is a medical context, we decided to emphasize false negatives, as they can lead to patients being left untreated. But if we only focused on that, that model would then diagnose more people with cancer to ensure no one is missed. To fix this, we used a portion of the malignant and benign recall scores from a default 60–40 split to obtain a more balanced model. This weighted recall metric was used to select the best model.

4 Naive Bayes



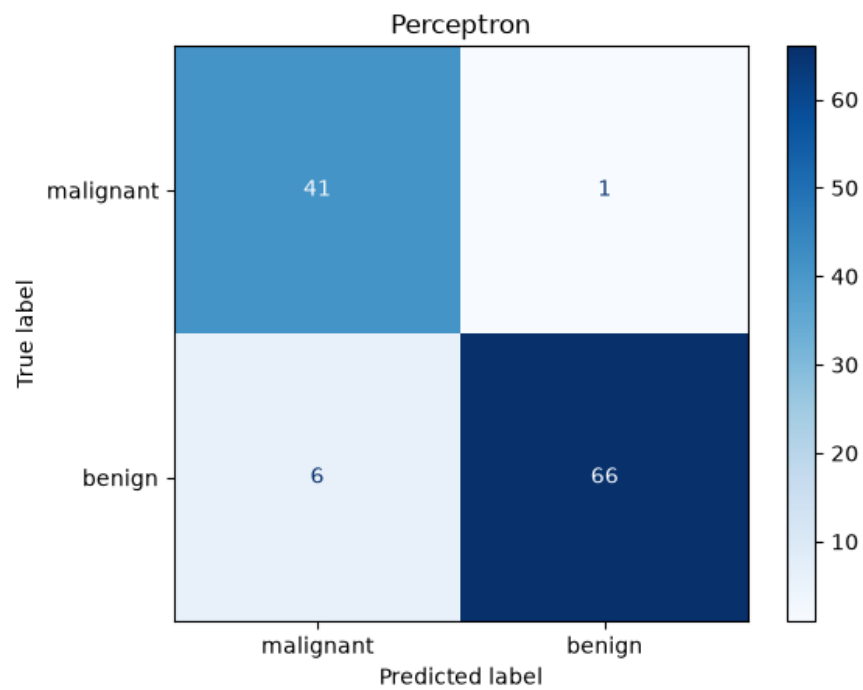
Confusion matrix for Naive Bayes model.

5 Logistic Regression



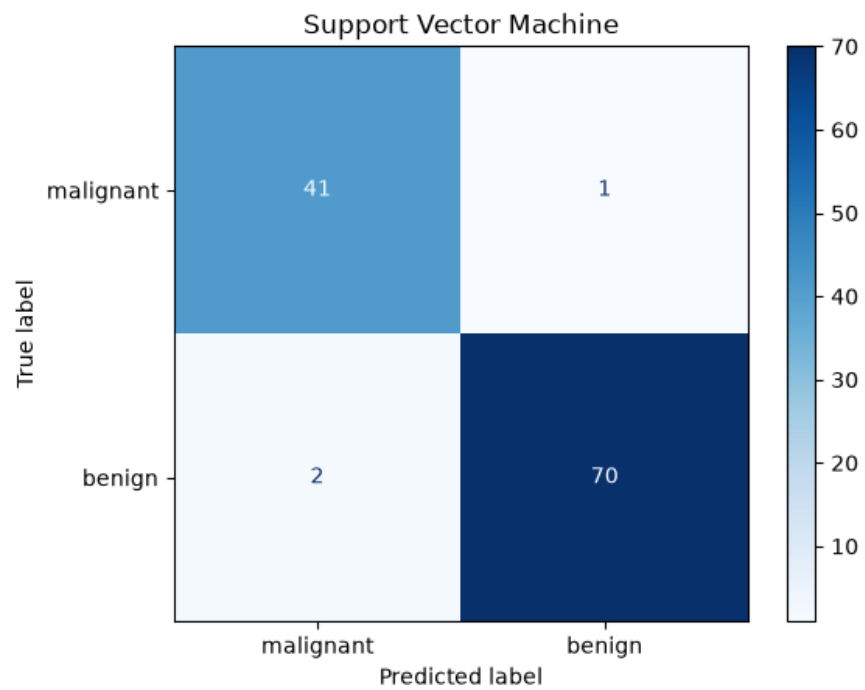
Confusion matrix for Logistic Regression model.

6 Perceptron



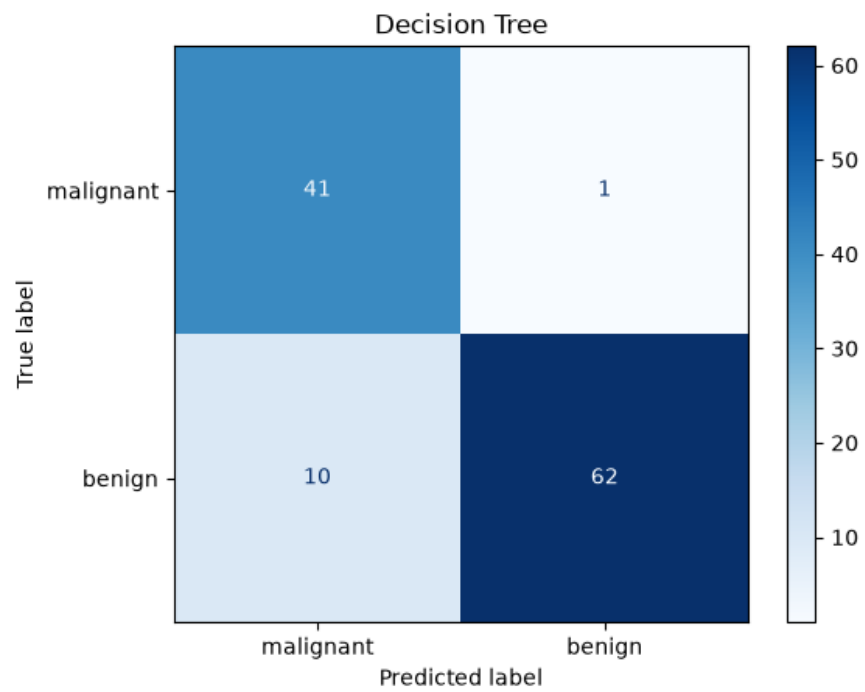
Confusion matrix for Perceptron model.

7 Support Vector Machine



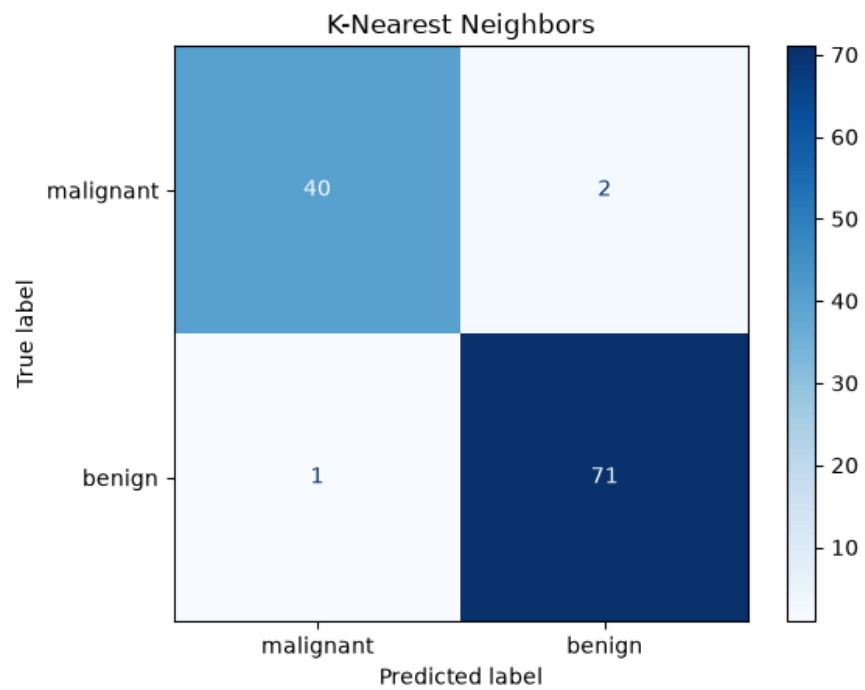
Confusion matrix for Support Vector Machine model.

8 Decision Tree Classifier



Confusion matrix for Decision Tree Classifier model.

9 K-Nearest Neighbors



Confusion matrix for K-Nearest Neighbors model.